Chapter 8. Classification Errors and the Bayes Classifier

Lecture 7-1 (error rate, Bayes optimal classifier, KNN, bias–variance, confusion matrix, ROC/AUC, CV for classification)

Contents

1 The Classification Problem **2**

2 The Bayes Optimal Classifier **2**

3 Approximating the Bayes Classifier **2**

3.1 K-nearest neighbors . . . 3

3.2 Bias–variance in classification . . . 3

4 Other Measures for Classification **3**

4.1 Confusion matrix . . . 3

4.2 Thresholds . . . 4

4.3 ROC curve and AUC . . . 4

5 Cross-Validation for Classification **4**

6 Selected exercises **5**

1 The Classification Problem

For regression we assessed accuracy by MSE; for classification we use the **error rate**

$$\text{Error rate} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(y_i \neq \hat{y}_i),$$

the proportion of misclassifications. Goals: build a classifier $C(X)$ that assigns a class label from $\mathcal{C}$ to a future unlabelled observation; assess the uncertainty of each classification; understand the roles of the predictors.

2 The Bayes Optimal Classifier

Motivation. The best possible classifier assigns each x to the most probable class; its error rate is a lower bound for every method. Classifiers can be understood as different ways of estimating these class probabilities.

Let the K classes be numbered $1, \ldots, K$. We want a classifier with small misclassification risk $R(C) = \Pr(C(X) \neq Y)$.

Theorem: Bayes classifier

The risk is minimized by

$$C^*(x) = \arg\max_k \Pr(Y = k \mid X = x),$$

assigning each x to the class with the largest conditional probability. Its risk $R(C^*) = 1 - \mathbb{E}\big[\max_k \Pr(Y = k \mid X)\big]$ is the **Bayes error rate**, the lowest error achievable by any classifier.

Proof [Supplement]. $R(C) = 1 - \Pr(C(X) = Y) = 1 - \mathbb{E}\big[\Pr(Y = C(X) \mid X)\big]$. For each x, $\Pr(Y = C(x) \mid X = x) \leq \max_k \Pr(Y = k \mid X = x)$, with equality when $C(x) = C^*(x)$. Taking expectations, $R(C) \geq R(C^*)$. ■

The Bayes error is what we would get if we knew the true distribution; it is positive whenever the classes overlap. In a simulated two-class example (100 observations per class), the **Bayes decision boundary** $\{x : \Pr(Y = 1 \mid x) = 0.5\}$ is a curved line.

3 Approximating the Bayes Classifier

In finite samples we measure a classifier $\hat{C}$ by its test error,

$$\text{Err}_{\text{Te}} = \text{Ave}_{i \in \text{Te}}\, \mathbb{I}\{y_i \neq \hat{C}(x_i)\}.$$

The Bayes classifier has the smallest population error; a good classifier has small test error. Two strategies: *estimate the conditional probabilities* and plug them into the Bayes rule (logistic regression, GAMs, LDA/QDA, KNN), or *build structured models for $C(x)$ directly* (support-vector machines).

3.1 K-nearest neighbors

KNN is a flexible approach to estimate the Bayes classifier. For a new point x, look at its neighbors:

$$\hat{\Pr}(Y = k \mid X = x) = \frac{1}{K} \sum_{x_i \in \mathcal{N}_K(x)} \mathbb{I}(y_i = k),$$

and predict the majority class among the K closest training points (for two classes coded 0/1, $\hat{f}(x) = \frac{1}{K}\sum y_i$ and classify by $\hat{f} > 0.5$). K controls complexity: **the smaller K, the more complex** the model. With $K = 1$ the boundary is overly flexible (training error 0); with $K = 100$ it is nearly linear and not flexible enough; $K = 10$ is close to the Bayes boundary.

3.2 Bias–variance in classification

As complexity increases ($1/K$ grows), training error always declines, while test error first declines (bias reduction dominates) and then increases (variance dominates): a U-shape. More flexible is not always better.

Remark: a classic result

As $n \to \infty$, the error of the 1-nearest-neighbor rule is at most twice the Bayes error (Cover and Hart, 1967): for two classes, $R_{1\text{NN}} \to \mathbb{E}[2\eta(X)(1-\eta(X))] \leq 2R^*(1-R^*)$ with $\eta(x) = \Pr(Y = 1 \mid x)$.

4 Other Measures for Classification

Motivation. Overall error rate hides the kinds of errors made; with rare classes it can look excellent for a useless classifier. Sensitivity, specificity and the ROC curve describe performance across all thresholds.

4.1 Confusion matrix

Default data (LDA fit, $n = 10{,}000$):

		True default status		
		No	Yes	Total
Predicted	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

- Overall error: $(23 + 252)/10000 = 2.75\%$ — but always predicting "No" would already give $333/10000 = 3.33\%$.
- **False positive rate**: fraction of negatives classified positive, $23/9667 = 0.2\%$.
- **False negative rate**: fraction of positives classified negative, $252/333 = 75.7\%$.

Sensitivity (recall, true positive rate) $= 1-\text{FNR} = 81/333 = 24.3\%$; specificity $= 1-\text{FPR} = 99.8\%$. A bank that wants to catch defaulters finds this unacceptable even though the overall error is low.

4.2 Thresholds

Classifying "Yes" when $\hat{\Pr}(\text{Yes} \mid x) \geq 0.5$ mimics the Bayes rule for overall error. Lowering the threshold (e.g. to 0.2 or 0.1) reduces the false negative rate at the cost of more false positives and possibly a higher overall error. The choice depends on the costs of the two errors: with costs c_{FN} and c_{FP}, predict "Yes" iff $\hat{p}\, c_{FN} > (1-\hat{p}) c_{FP}$, i.e. $\hat{p} > c_{FP}/(c_{FP}+c_{FN})$.

4.3 ROC curve and AUC

The **ROC curve** plots the true positive rate against the false positive rate as the threshold varies from 1 to 0, displaying both error types simultaneously. The **AUC** (area under the curve) summarizes overall performance; higher is better, 0.5 is random guessing and 1 is perfect separation.

Theorem: probabilistic meaning of AUC

For a score $s(x)$, $\text{AUC} = \Pr\left(s(X^+) > s(X^-)\right)$ for independent draws X^+ from the positive and X^- from the negative class (ties counted 1/2).

Proof [Supplement]. With threshold t, $\text{TPR}(t) = \Pr(s^+ > t)$ and $\text{FPR}(t) = \Pr(s^- > t)$. Then $\text{AUC} = \int \text{TPR}\, d\text{FPR} = \int \Pr(s^+ > t)\, dF_-(t) = \Pr(s^+ > s^-)$. ■

AUC depends only on the ranking of scores, so it is invariant to monotone transformations and to the class proportions.

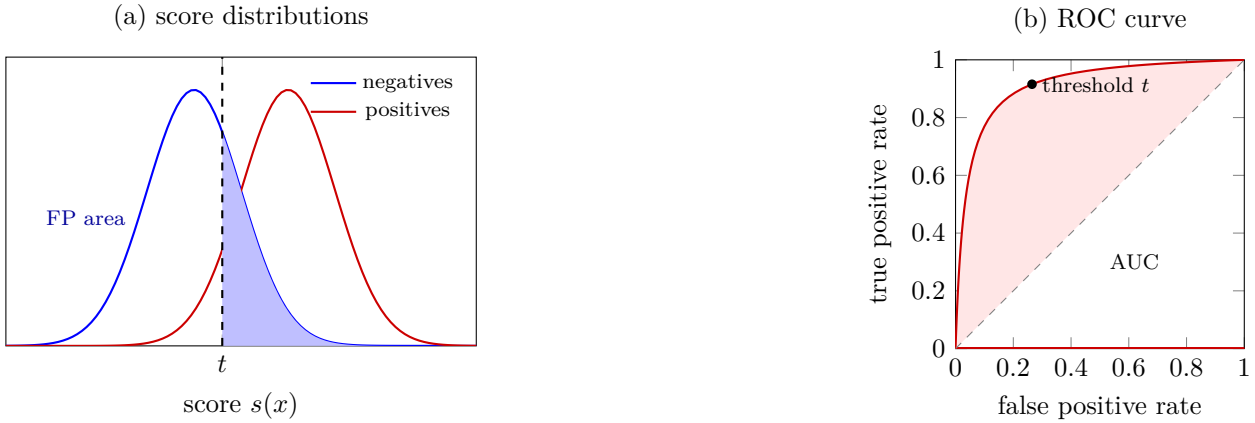


Figure 1: (a) Scores of negatives and positives; classifying "positive" above a threshold t gives a false positive rate (shaded) and a true positive rate. (b) Moving t from $+\infty$ to $-\infty$ traces the ROC curve; the area below it is the AUC. The diagonal is random guessing (AUC = 0.5).

5 Cross-Validation for Classification

Divide the data into K roughly equal parts $C_1, \ldots, C_K$ with n_k observations in part k and compute

$$\text{CV}_K = \sum_{k=1}^{K} \frac{n_k}{n} \text{Err}_k, \qquad \text{Err}_k = \frac{1}{n_k} \sum_{i \in C_k} \mathbb{I}(y_i \neq \hat{y}_i),$$

with $\hat{y}_i$ from the model fitted without part k. On two-dimensional simulated data, for logistic regression with polynomial predictors of increasing order and for KNN with decreasing K, the 10-fold CV curve tracks the true test error closely (it may underestimate the level, but its minimum is near the right complexity), while the training error keeps falling.

Summary

- The Bayes classifier $\arg\max_k \Pr(Y = k \mid x)$ minimizes the error rate; its error is a lower bound.
- KNN estimates the conditional probabilities by local voting; small K is flexible, large K smooth.
- The overall error rate can hide poor performance on a rare class; report FPR, FNR, sensitivity and specificity, and choose thresholds by costs.
- ROC curves show all thresholds; AUC is the probability of ranking a random positive above a random negative.
- CV with the misclassification loss selects complexity for classifiers.

6 Selected exercises

Motivation. Three results stated in this chapter without proof: the cost-weighted threshold, the algebra of the Cover–Hart bound, and the Bayes error of two Gaussian classes quoted in the LDA simulation of Ch. 10.

Exercise 8.1 (The cost-sensitive Bayes rule)

Let $Y \in \{0, 1\}$, $\eta(x) = \Pr(Y = 1 \mid X = x)$, and let a classifier C incur cost $c_{FN} > 0$ when $C(X) = 0$, $Y = 1$ and $c_{FP} > 0$ when $C(X) = 1$, $Y = 0$. Show that the expected cost

$$R(C) = c_{FN} \Pr(C(X) = 0, Y = 1) + c_{FP} \Pr(C(X) = 1, Y = 0)$$

is minimized by $C^*(x) = \mathbb{I}\{\eta(x) > c_{FP}/(c_{FP} + c_{FN})\}$, which is the Bayes classifier when $c_{FP} = c_{FN}$.

Solution [Supplement].

Step 1. Conditioning on X,

$$R(C) = \mathbb{E}\big[c_{FN}\, \eta(X) \mathbb{I}\{C(X) = 0\} + c_{FP}\{1 - \eta(X)\}\mathbb{I}\{C(X) = 1\}\big].$$

Step 2. For each x the integrand is $c_{FN}\eta(x)$ if $C(x) = 0$ and $c_{FP}\{1 - \eta(x)\}$ if $C(x) = 1$, so it is at least $\min\{c_{FN}\eta(x), c_{FP}(1 - \eta(x))\}$, the value attained by C^*: $C^*(x) = 1$ exactly when $c_{FP}(1 - \eta) < c_{FN}\eta$, i.e. $\eta > c_{FP}/(c_{FP} + c_{FN})$ (at equality both choices cost the same).

Step 3. Taking expectations, $R(C) \geq R(C^*)$. With equal costs the threshold is $1/2$. If missing a defaulter costs nine times a false alarm, the threshold is 0.1, the value used for the Default data in Ch. 10. ■

Exercise 8.2 (The Cover–Hart bound)

Let $\eta(x) = \Pr(Y = 1 \mid X = x)$ and $R^* = \mathbb{E}\min\{\eta(X), 1 - \eta(X)\}$ the Bayes error. Show that

$$R^* \leq \mathbb{E}[2\eta(X)\{1 - \eta(X)\}] \leq 2R^*(1 - R^*) \leq 2R^*.$$

Solution [Supplement]. Let $M = \min\{\eta(X), 1 - \eta(X)\} \in [0, \frac{1}{2}]$. Since $\{\eta, 1 - \eta\} = \{M, 1 - M\}$, $\eta(1 - \eta) = M(1 - M)$.

Step 1. $2(1-M) \geq 1$ because $M \leq \frac{1}{2}$, so $2M(1-M) \geq M$; taking expectations gives the first inequality.

Step 2. By Jensen's inequality $\mathbb{E}M^2 \geq (\mathbb{E}M)^2 = R^{*2}$, hence $\mathbb{E}[2M(1-M)] = 2(R^* - \mathbb{E}M^2) \leq 2R^*(1-R^*)$.

Step 3. $2R^*(1-R^*) \leq 2R^*$ trivially.

The middle quantity is the large-sample error of 1-NN: the nearest neighbor's label behaves like an independent draw with the same η, and the two labels disagree with probability $\eta(1-\eta)+(1-\eta)\eta$. So 1-NN is never better than Bayes and at most twice as bad; for $R^* = 0.1$ its limiting error is at most 0.18. ■

Exercise 8.3 (Bayes error of two Gaussian classes)

Let $\pi_1 = \pi_2 = \frac{1}{2}$ and $X \mid Y = k \sim N(\mu_k, \Sigma)$ in $\mathbb{R}^p$, $\Sigma \succ 0$, $\mu_1 \neq \mu_2$, and $\Delta^2 = (\mu_2 - \mu_1)^\top \Sigma^{-1}(\mu_2 - \mu_1)$ (Mahalanobis distance).
(a) Show that the Bayes classifier assigns class 2 iff $w^\top x > w^\top(\mu_1+\mu_2)/2$, with $w = \Sigma^{-1}(\mu_2 - \mu_1)$.
(b) Show that the Bayes error is $\Phi(-\Delta/2)$. Check the value $\Phi(-1.5) \approx 6.7\%$ for $p = 1$, $\mu_{1,2} = \mp 1.5$, $\sigma = 1$.

Solution [Supplement].

(a) With equal priors the Bayes rule picks the larger density. The normalizing constants are equal, so

$$\log \frac{f_2(x)}{f_1(x)} = \frac{1}{2}\{(x-\mu_1)^\top\Sigma^{-1}(x-\mu_1) - (x-\mu_2)^\top\Sigma^{-1}(x-\mu_2)\} = w^\top x - \frac{1}{2}w^\top(\mu_1+\mu_2),$$

where the terms $x^\top\Sigma^{-1}x$ cancel and $\mu_2^\top\Sigma^{-1}\mu_2 - \mu_1^\top\Sigma^{-1}\mu_1 = (\mu_1+\mu_2)^\top\Sigma^{-1}(\mu_2-\mu_1)$ by symmetry of Σ^{-1}. Class 2 wins iff this is positive.

(b)

Step 1. Put $c = w^\top(\mu_1+\mu_2)/2$. Under class 1, $w^\top X \sim N(w^\top\mu_1, w^\top\Sigma w)$ with $w^\top\Sigma w = \Delta^2$, and $c - w^\top\mu_1 = \frac{1}{2}w^\top(\mu_2-\mu_1) = \Delta^2/2$. Hence $\Pr(w^\top X > c \mid Y=1) = 1 - \Phi(\Delta/2) = \Phi(-\Delta/2)$.

Step 2. Under class 2, $w^\top\mu_2 - c = \Delta^2/2$, so $\Pr(w^\top X \leq c \mid Y=2) = \Phi(-\Delta/2)$.

Step 3. The Bayes error is $\frac{1}{2}\Phi(-\Delta/2) + \frac{1}{2}\Phi(-\Delta/2) = \Phi(-\Delta/2)$. For $p=1$, $\Delta = |\mu_2-\mu_1|/\sigma = 3$, giving $\Phi(-1.5) = 0.0668$. The error depends on the classes only through the Mahalanobis distance, not on the raw distance $\|\mu_2 - \mu_1\|$. ■